

Realistic Challenges and Optimization Strategies for High-Quality Development of Cross-Border E-Commerce in Xinjiang, China in the Context of Digital Technology

Cui Mengyuan¹, Yang Tianxiang²

¹ School of International Economics and Trade, Xinjiang University of Finance and Economics; China; 825658119@qq.com

² School of International Economics and Trade, Xinjiang University of Finance and Economics; China; 2067438520@qq.com

Abstract: Against the backdrop of the vigorous development of digital technology, this study depicts the foundational support for the high-quality development of cross-border e-commerce in Xinjiang by presenting operational data and institutional practices of cross-border e-commerce in 6 major node cities of Xinjiang from 2024 to 2025, as well as the current status of infrastructure and industrial foundation in the region. It further analyzes realistic challenges restricting the empowerment effect of digital technology from three dimensions: the depth of digital technology application, policy coordination and talent supply, and industrial ecological synergy. The research shows that cross-border e-commerce in Xinjiang has achieved explosive growth driven by digital technology and policy dividends, yet there exists a structural contradiction of "strong hardware yet weak software". On this basis, this study proposes a three-dimensional collaborative optimization strategy of "digital infrastructure – policy system – industrial integration" to address the current realistic constraints on the empowerment effect of digital technology, so as to provide decision-making references for Xinjiang to build a highland for inland and border opening-up.

¹ Corresponding Author: Cui Mengyuan (825658119@qq.com)

² Corresponding Author: Yang tianxiang(2067438520@qq.com)

Keywords: Xinjiang; digital technology; cross-border e-commerce; optimization strategies

1.Introduction

1.1 Research Background

Against the backdrop of profound adjustments in the global economic landscape and the surging wave of digitalization, the digital economy has become a key force reshaping global value chains and transforming the pattern of international trade division. As a product of the in-depth integration of digital technology and international trade in the new era, cross-border e-commerce is emerging as an important engine for building the new development pattern of dual domestic and international circulation. As the core area of the Silk Road Economic Belt, Xinjiang enjoys unique geographical advantages connecting the Eurasian market. With the in-depth advancement of the Belt and Road Initiative, how to use digital technology to break geographical space constraints, overcome the natural disadvantage of traditional inland border areas being far from central markets, and achieve leapfrog development of the foreign trade industry is a major historical issue facing Xinjiang ^[1-3]. In recent years, cross-border e-commerce in Xinjiang has shown explosive growth. According to measured data from 2024 to 2025, the import and export volume of cross-border e-commerce in Urumqi, the capital city, has exceeded 8.3 billion yuan, with a year-on-year increase of 222%, accounting for a considerable proportion of the city's total import and export value. Alashankou, as a port hub, has handled over 55 million export parcels cumulatively, with a value approaching 4 billion yuan, achieving nearly doubled growth. Behind these remarkable data is the deep penetration and wide application of digital technology in the entire trade chain. At present, the construction of digital infrastructure in Xinjiang has achieved phased results. The region has built more than 15,000 5G base stations cumulatively, with 80% 5G coverage in administrative villages. At the micro-enterprise level, the penetration rate of ERP and

purchase-sales-inventory management software in enterprises above designated size has gradually increased. Digital applications such as intelligent logistics and cloud warehousing are reconstructing traditional logistics operation modes and significantly improving cargo turnover efficiency ^[4]. However, despite the rapid development momentum, the high-quality development of cross-border e-commerce in Xinjiang still faces many deep-seated constraints. Compared with developed eastern coastal areas, Xinjiang still has shortcomings such as long logistics chains, high comprehensive trade costs and insufficient industrial digital supporting facilities. With the State Council's approval of the establishment of new cross-border e-commerce comprehensive pilot zones such as Bortala Prefecture, and the implementation of a series of trade facilitation innovation systems such as "intelligent image review" and "7×24-hour customs clearance", the interactive relationship between policy supply and technology application has become increasingly complex and important. This study deeply explores the paths and countermeasures of digital technology empowering the high-quality development of cross-border e-commerce in Xinjiang, which has theoretical value and practical guiding significance.

1.2 Domestic and Foreign Research Status

Existing studies have formed multi-dimensional achievements focusing on the mechanism of digital technology empowering international trade, performance evaluation and influencing factors of cross-border e-commerce development, institutional environment and policy adjustment effects, and open digital transformation in Xinjiang and border areas.

In terms of research on the mechanism of digital technology empowering international trade, Freund and Weinhold (2004) early used the gravity model to confirm a significant positive correlation between internet penetration and service trade exports, pointing out that network technology effectively reduces market entry barriers ^[5]. Goldfarb and Tucker (2019) pointed out that digital technology not only reduces search costs and replication costs, but also changes the generation logic of global value chains through the flow of data factors^[6].

In terms of research on cross-border e-commerce development performance evaluation and influencing factors, OECD (2019) proposed a comprehensive framework for digital trade ^[7]. Li Qi et al. (2016) constructed China's cross-border e-commerce development index ^[8]. Regarding influencing factors, Terzi (2011) emphasized the importance of the institutional environment, arguing that customs clearance efficiency and the perfection of laws and regulations are the key to determining a country's cross-border e-commerce competitiveness^[9]. Yue Yunsong et al. (2020) confirmed that the establishment of comprehensive pilot zones significantly promotes urban export growth, with a stronger effect on remote areas^[10].

In terms of research on the institutional environment and policy adjustment effects, Acemoglu (2005) emphasized that institutions determine the adoption efficiency and incentive mechanism of technology ^[11]. Ma Shuzhong et al. (2019) proposed that institutional innovations such as comprehensive pilot zones effectively reduce policy uncertainty by providing a "regulatory sandbox" ^[12].

In terms of research on open digital transformation in Xinjiang and border areas, Gao Zhigang et al. (2014) discussed Xinjiang's hub position under the background of the Silk Road Economic Belt and argued that infrastructure connectivity is the core task ^[13]. Sun Zhenyan (2022) analyzed the development status of cross-border e-commerce in Xinjiang and put forward suggestions to improve customs clearance facilitation based on digital technology ^[14].

Existing research results have laid an important foundation for understanding the empowerment effect of digital technology and the institutional environment of cross-border e-commerce. However, for Xinjiang as a typical inland border area, there is insufficient analysis of the factors restricting the insufficient empowerment effect of digital technology. To address this gap, this study focuses on the foundational support for the high-quality development of cross-border e-commerce in Xinjiang, analyzes the factors constraining the release of digital technology's empowerment effect, and proposes targeted optimization strategies accordingly.

2. Foundational Support for the High-Quality Development of Cross-Border E-Commerce in Xinjiang

2.1 Resource Foundation

Xinjiang is located in the hinterland of the Eurasian continent, bordering 8 countries. Land transportation saves two-thirds of the time compared with sea transportation and four-fifths of the cost compared with air transportation, and it has 17 national first-class ports.

In terms of trade stock, in the first half of 2025, the import and export volume of cross-border e-commerce in Urumqi reached 8.31 billion yuan (accounting for 16% of the city's total import and export value); Alashankou handled 55.9819 million export parcels cumulatively (valued at 3.94 billion yuan); Kashgar achieved 7 billion yuan in orders relying on the "Global Trade Connect" platform. In addition, Xinjiang's characteristic industries, including ten industrial clusters such as cotton, textile and garment, and new energy and new materials (286,900 enterprises), are also key elements supporting the development of cross-border e-commerce. To intuitively reflect the actual level and differences of cross-border e-commerce development in different regions of Xinjiang, this study compiles the core business data of major node cities from 2024 to 2025, as shown in Table 1. It can be seen from Table 1 that Urumqi and Alashankou, as the core of comprehensive pilot zones, are in a leading position in both trade scale and growth rate.

Table 1. Comparison of Key Indicators of Cross-Border E-Commerce Development in Major Node Cities of Xinjiang (2024-2025)

City	CPZ Status	Import & Export Volume in H1 2025 (100 million yuan)	Year-on-year Growth Rate (%)	Export Parcel Volume (10,000 parcels)	Number of 5G Base Stations	Core Advantages
------	------------	--	------------------------------	---------------------------------------	----------------------------	-----------------

Urumqi	yes	83.10	222.0%	45.00*	8500+	Full-mode operation, cargo distribution center
Alashankou	yes	35.79	103.0%	5598.19	1200+	Channel logistics, smart port
Kashgar	yes	70.00*	56.5%	1230.50	2100+	Platform-driven, southern Xinjiang pivot
Khorgos	no	28.50	45.2%	850.20	800+	Border trade, highway-rail intermodal transport
Karamay	no	15.60	32.8%	320.10	1500+	Smart logistics, digital transformation
Bortala Prefect	yes (new	12.30	88.6%	560.00	900+	Policy lowland, port

ure	ly	linkage
	approv	
	ed)	

**Note: The parcel volume of Urumqi is data from the airport port; Kashgar data includes platform order volume.*

2.2 Policy Foundation

At present, Xinjiang has formed a policy superimposition advantage with the cross-border e-commerce comprehensive pilot zone (hereinafter referred to as CPZ) as the core carrier and a series of customs clearance and trade facilitation innovation systems as specific starting points, laying a solid institutional foundation for the leapfrog development of cross-border e-commerce. As shown in Figure 1, driven by both digital technology and comprehensive pilot zone policies, cross-border e-commerce in Xinjiang has shown explosive growth.

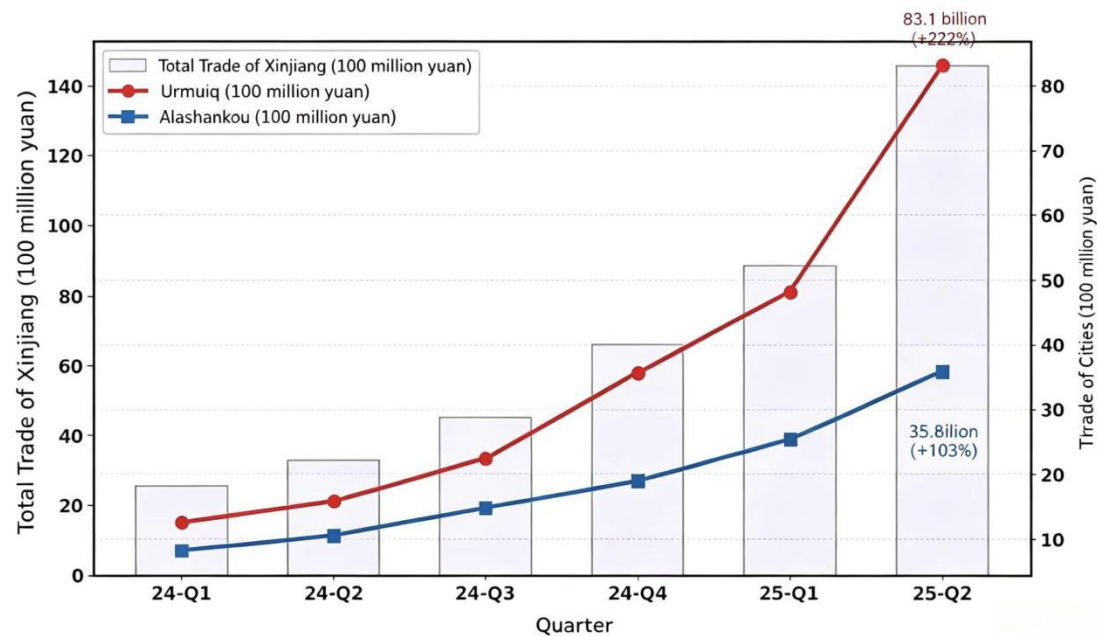


Figure 1 .Growth Trend of Cross-Border E-Commerce Trade Volume of Major Cities in Xinjiang, 2024-2025

First, in recent years, Xinjiang has formed a CPZ matrix of "multi-point support and whole-region linkage" consisting of Urumqi, Alashankou, Kashgar and Bortala

Prefecture, with policies such as tax exemption for retail exports and verified collection of enterprise income tax. Second, the customs has implemented innovative systems such as "7×24-hour customs clearance" (multiple times improvement in efficiency), "intelligent image review" and batch inspection mode (20-fold improvement in inspection efficiency), and the mode of "list verification and release, summary statistics". Third, local governments have introduced precise gradient reward mechanisms (e.g., enterprises in Alashankou with a cargo value exceeding 10 million yuan are rewarded 0.01 yuan for every 6 yuan of cargo value) and financial support such as the launch of the "Cross-border Payment Connect" service (second-level arrival of cross-border receipts).

2.3 Infrastructure Foundation

At present, through large-scale investment in "new infrastructure" and digital transformation of traditional logistics channels, Xinjiang is building an infrastructure network with deep integration of the "sky network" (digital network) and the "ground network" (logistics network).

In terms of "sky network" construction, the region has built more than 15,000 5G base stations cumulatively in recent years, and the Urumqi International Land Port Area has deployed an industrial-grade private network based on 5G technology. In terms of "ground network" construction, a three-dimensional logistics channel network has been formed with China-Europe Railway Express as the backbone and highway transportation as capillaries. The construction of automated logistics management systems in Karamay Smart Logistics Park has improved cargo turnover efficiency by more than 30%. Alashankou and Khorgos have advanced the automatic transformation of port transshipment facilities and laid out a warehousing network featuring "domestic cargo collection warehouses + border transit warehouses + overseas warehouses".

2.4 Industrial Foundation

In recent years, driven by both digital technology and policy dividends, the industrial foundation of cross-border e-commerce in Xinjiang has undergone profound changes, as detailed below.

According to the latest statistics, the penetration rate of information management systems in "four above-scale" enterprises reaches 97.2%, and the ERP penetration rate of enterprises above designated size stands at 35.6%. Diversified market entities are flourishing: leading platforms such as Alibaba International Station and JD International have entered Xinjiang, and local platforms such as "Meilingou" and "Global Trade Connect" have emerged. By the end of June 2025, there were 209 enterprises engaged in cross-border e-commerce in Urumqi, covering the entire industrial chain including e-commerce platform operation, supply chain services, logistics and distribution, marketing promotion, cross-border customs clearance, and incubation and cultivation.

3. Realistic Challenges Restricting the Empowerment Effect of Digital Technology

Although cross-border e-commerce in Xinjiang has achieved remarkable results driven by digital technology, the full play of digital technology's empowerment effect faces multiple bottleneck constraints, concentrated in the structural contradiction of "strong hardware yet weak software" — hardware infrastructure construction has progressed rapidly, but there are obvious shortcomings at the software level in terms of technology application depth, policy coordination and talent support, and industrial supporting facilities.

3.1 Insufficient Depth of Digital Technology Application

At present, the construction of digital infrastructure in Xinjiang has made phased breakthroughs, with the number of 5G base stations growing continuously and new network facilities such as SD-WAN dedicated lines put into use successively.

However, the in-depth application of digital technology still lags significantly behind the pace of hardware construction, which is prominently reflected in that the digital processing capacity of logistics hubs and port nodes is far from meeting the requirements of efficient operation. Specifically, on the one hand, although core hubs such as Urumqi International Land Port Area, Alashankou and Khorgos have achieved 5G network coverage, network performance in terms of latency and connection density cannot yet meet the demand for large-scale automated operations when facing scenarios such as simultaneous access of massive IoT devices and high-frequency real-time data backhaul. On the other hand, the allocation rate of digital equipment such as intelligent sorting lines, automated stereoscopic warehouses and full-process cargo tracking systems in ports and logistics parks is relatively low, and a large number of operations still rely on manual work, with data flow not yet interconnected among nodes. This disconnect between "hardware" and "software" greatly weakens the empowerment effect of digital technology on cross-border logistics efficiency. In addition, the penetration rate of digital management tools at the enterprise level remains low. Only 41.67% of e-commerce platforms have built data middle platforms, and a large number of small and medium-sized enterprises have not yet completed the digital transformation of business processes. This means that although more than half of platform enterprises possess business data, such data often lies dormant in fragmented systems such as orders, inventory and logistics, failing to be cleaned, integrated and standardized, let alone transformed into decision-supporting business intelligence (BI). The core of digital technology empowerment lies in converting data factors into productivity. However, at present, cross-border e-commerce enterprises in Xinjiang are still in the exploratory stage in high-end application scenarios such as precision marketing, intelligent customer service and supply chain forecasting, and the mining of data value is far from sufficient. This tendency of "emphasizing construction while neglecting application" prevents a large amount of hardware investment from being translated into corresponding efficiency improvement and competitiveness enhancement.

3.2 Insufficient Policy Coordination and Talent Shortage

Although Xinjiang has formed a CPZ policy matrix covering 4 zones, the coordinated linkage among various CPZs is still insufficient, and the radiation-driven effect of policy dividends needs to be further released. Cities with active cross-border e-commerce such as Korgos have not yet been included in the CPZ scope, resulting in policy gaps in enjoying customs clearance facilitation and tax incentives. Furthermore, the structural shortage of compound digital talents is a key pain point restricting digital technology from empowering the high-quality development of cross-border e-commerce in Xinjiang. Technology is the tool of empowerment, while talents are the subject of empowerment. As an inland region, Xinjiang has long faced the dilemma of difficulty in introducing and retaining talents, which is particularly prominent in the cross-border e-commerce field. Cross-border e-commerce enterprises in Xinjiang have an average of only 2 IT technical personnel. Such stretched staffing makes most enterprises only able to maintain daily operation and maintenance of basic systems, and incapable of carrying out high-end technology research and development such as algorithm optimization and big data mining. More seriously, there is an extreme shortage of compound talents proficient in foreign languages, trade and digital operation. Against the booming development of new formats such as live-streaming e-commerce and short-video marketing, the lack of professional MCN institutions and operation specialists makes it difficult for many high-quality characteristic products in Xinjiang to tell brand stories through digital means, resulting in weak traffic monetization capability. The shortage of talent factors has become the biggest constraint restricting the transformation of cross-border e-commerce in Xinjiang from "extensive growth" to "refined operation", and may even cause expensive digital infrastructure to fall into "idleness" due to lack of operation.

3.3 Insufficient Industrial Ecological Synergy

Digital technology empowerment emphasizes ecological synergy, which requires close cooperation of upstream and downstream industrial chains. However, the

current industrial ecosystem of cross-border e-commerce in Xinjiang is still immature, presenting a pattern of "strong in key nodes but weak in overall coverage". Although core nodes such as Urumqi and Alashankou develop rapidly, the vast hinterland and surrounding industrial belts have not yet achieved in-depth digital connection. Many traditional manufacturing enterprises (such as textile and garment, agricultural product processing) still have a low level of digitalization and insufficient ERP penetration, making it difficult to match the flexible supply chain requirements of cross-border e-commerce platforms. In addition, although the number of service providers such as cross-border logistics, payment and settlement, and overseas warehousing has increased, their overall scale is small and service standards are inconsistent, with no leading enterprises with international competitiveness yet formed. This "temperature difference" in the digitalization level of upstream and downstream industrial chains leads to the loss of collaborative efficiency. For example, when an e-commerce platform receives a large overseas order, upstream factories often fail to deliver on time due to lack of digital production scheduling capability, and such supply-demand mismatch weakens the agility of the overall supply chain. In addition, small and medium-sized cross-border e-commerce enterprises generally face the problem of difficult and expensive financing. The traditional credit model can hardly convert enterprises' operational data such as customs clearance, logistics and payment into credit assets, and financial services provide insufficient support for the industrial ecosystem. How to effectively transform "trade flow" into "industrial retention" is a problem that must be solved for cross-border e-commerce in Xinjiang to move from scale expansion to high-quality development.

4. Optimization Strategies for Digital Technology

Empowering Cross-Border E-Commerce Development in Xinjiang

In view of the current structural contradiction of "strong hardware yet weak software", this chapter proposes a three-dimensional collaborative optimization strategy of "digital infrastructure – policy system – industrial integration", aiming to provide a scientific action guide for Xinjiang to build a highland for inland and border opening-up.

4.1 Consolidating the Digital Foundation: Strengthening New Infrastructure and Data Governance Capabilities

First, promote the iterative upgrading of digital infrastructure. Accelerate the deployment of 5G-A (5.5G) base stations to achieve full coverage of millisecond-level low-latency networks in core industrial belts such as Changji Prefecture and Urumqi International Land Port Area. Meanwhile, support ports and logistics hubs to introduce automatic sorting lines, intelligent stereoscopic warehouses and logistics tracking equipment on a large scale to improve the digital processing capacity of logistics nodes. Second, implement the action of "enterprise cloud adoption, data utilization and intelligence empowerment". The government sets up special guiding funds and guides e-commerce platforms and leading enterprises to build industry-level data middle platforms, so as to reduce the cloud adoption cost of small and medium-sized enterprises and eliminate information islands. Meanwhile, encourage enterprises to apply digital management systems such as ERP and WMS to lay a micro foundation for industrial digital transformation.

4.2 Improving Institutional Supply: Amplifying Policy Dividends and Adjustment Effects of Comprehensive Pilot Zones

First, expand the coverage and spillover effects of CPZ policies. Actively strive to include active cross-border e-commerce regions such as Khorgos and Tacheng into the national CPZ scope to form a policy matrix with whole-region linkage. Promote

customs clearance modes such as "list verification and release, summary statistics" to further compress the inspection and release time of special categories. Meanwhile, establish a paired assistance mechanism between CPZs and non-CPZs to narrow the "digital divide" between regions. Second, establish a precise gradient reward and talent introduction and cultivation mechanism. Establish a more precise financial incentive mechanism, focusing on supporting enterprises' investment in key links such as digital system R&D and overseas warehouse construction. Rely on local universities in Xinjiang to set up micro-majors in cross-border e-commerce, and introduce flexible talent introduction policies to attract digital operation experts from eastern regions through "migratory bird-style employment", so as to make up for the shortage of talents and provide intellectual support for industrial development.

4.3 Building Industrial Ecology: Promoting the Integration of "Technology + Industry"

First, promote the integrated development of "cross-border e-commerce + industrial belts". Rely on ten industrial clusters such as cotton textile and new energy to cultivate vertical cross-border e-commerce platforms. Support traditional manufacturing enterprises to connect with international market demand through the C2M (Consumer-to-Manufacturer) mode, and promote "Xinjiang-made" products to transform towards branding and high added value. For example, use the platform data of Kashgar's "Global Trade Connect" to guide the textile and garment industry in southern Xinjiang to accurately connect with the demand of the Central Asian market, realizing seamless connection between production capacity and market. Second, deepen the integrated innovation of "digital technology + financial services". Further expand the coverage of the "Cross-border Payment Connect" service and introduce more banks and financial institutions to participate. Use blockchain technology to build a cross-border trade credit data chain, convert enterprises' customs clearance, logistics and payment data into credible credit assets, and develop pure credit financial products such as "cash flow loans" and "order-based loans". This can not only reduce financial risks, but also inject financial vitality into small, medium and

micro enterprises, and build a benign ecological closed loop of "technology supporting transactions, transactions precipitating data, data exchanging for credit, and credit obtaining financing".

References

- [1] Wang X, Fan Z L. AI reshapes the supply chain ecosystem of cross-border e-commerce [J]. *Enterprise Management*, 2025, (11): 112-115.
- [2] CNR. Export parcels of Alashankou cross-border e-commerce comprehensive pilot zone exceed 5 million [EB/OL]. (2025-02-20)[2025-08-10]. https://xj.cnr.cn/xjfw_1/tpxw/20250220/t20250220_527076314.shtml.
- [3] Chinanews.com. Trade volume of Kashgar Global Trade Connect platform reaches nearly 7 billion yuan [EB/OL]. (2025-08-05)[2025-08-10]. <http://www.xj.chinanews.com.cn/jingji/2025-08-05/detail-ihetykaf8558292.shtml>.
- [4] Wang L N, Hao Y Y, Jiang Y F. Spatiotemporal characteristics and driving mechanism of coupling coordination between China's digital economy and high-quality development of cross-border e-commerce [J]. *China Business & Trade*, 2025, 34(21): 41-46.
- [5] Freund C L, Weinhold D. The effect of the Internet on international trade [J]. *Journal of International Economics*, 2004, 62(1): 171-189.
- [6] Goldfarb A, Tucker C. Digital economics [J]. *Journal of Economic Literature*, 2019, 57(1): 3-43.
- [7] OECD. Measuring Digital Trade: Towards a Conceptual Framework [R]. OECD Trade Policy Papers, No. 227. Paris: OECD Publishing, 2019.
- [8] Li Q, Wei L B. Construction of evaluation system for cross-border e-commerce development index: An empirical study based on China's provincial data [J]. *E-Government*, 2016, (10): 12-21.
- [9] Terzi N. The impact of e-commerce on international trade and employment [J]. *Procedia - Social and Behavioral Sciences*, 2011, 24: 745-753.
- [10] Yue Y S, Li B, Li R. E-commerce platform application and export performance of Chinese manufacturing enterprises [J]. *The Journal of World Economy*, 2020, 43(10): 147-172.
- [11] Acemoglu D, Johnson S, Robinson J A. Institutions as a fundamental cause of long-run growth [A]. *Handbook of Economic Growth* [C]. 2005, 1: 385-472.

[12] Ma S Z, Fang C, Liang Y H. Digital trade: Conceptual connotation, development trend and policy suggestions [J]. International Economic Review, 2019, (05): 51-67+5.

[13] Gao Z G, Kezhu Latif. Construction of the Silk Road Economic Belt and strategic choice of Xinjiang's foreign trade development [J]. Journal of Xinjiang University (Philosophy, Humanities & Social Sciences), 2014, 42(03): 1-6.

[14] Sun Z Y. Research on the current situation, problems and countermeasures of cross-border e-commerce development in Xinjiang [J]. Foreign Economic Relations & Trade, 2022, (04): 18-21.